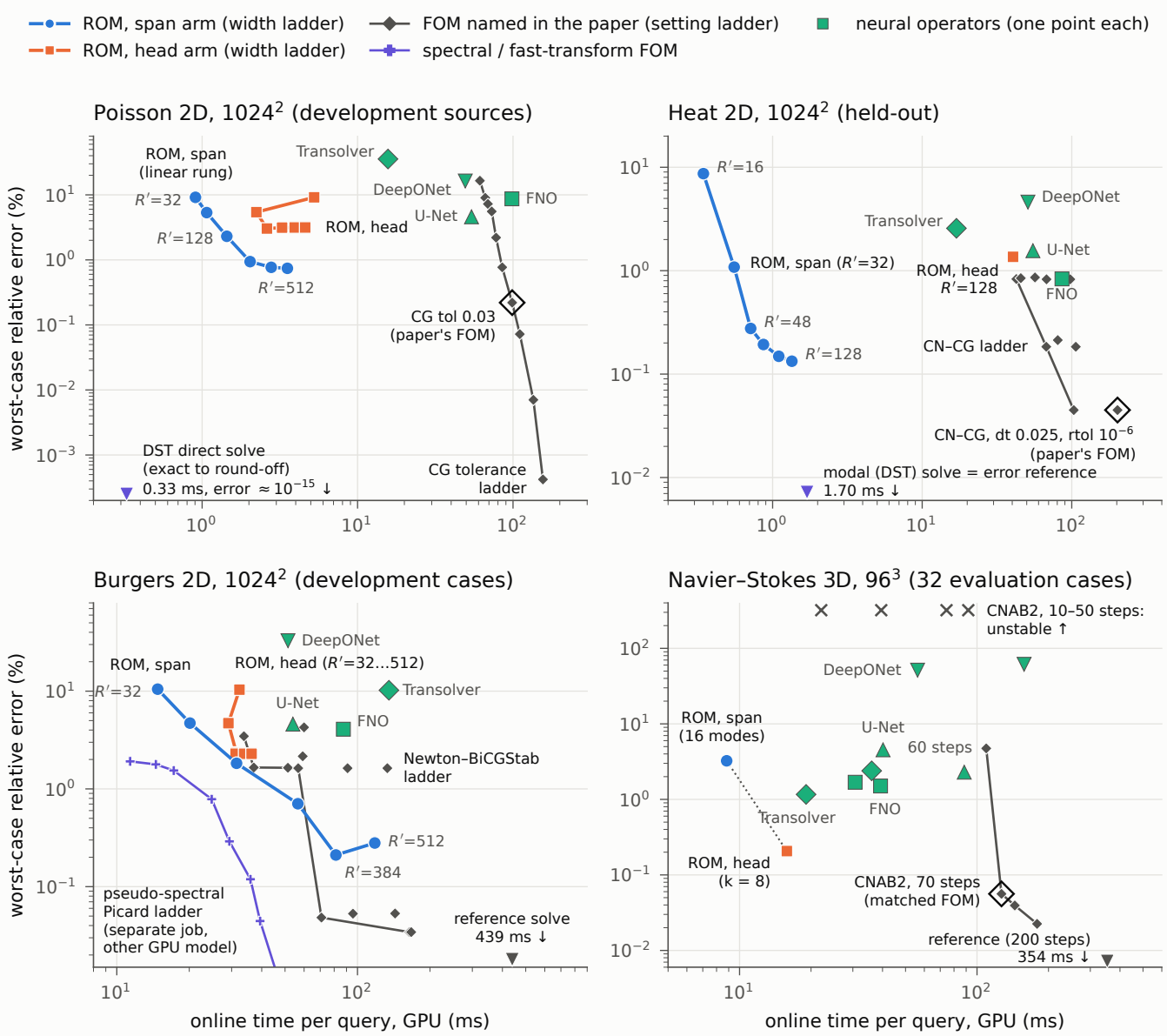
ROM, span arm (width ladder)
ROM, head arm (width ladder)
FOM named in the paper (setting ladder)
spectral / fast-transform FOM
neural operators (one point each)
Poisson 2D, 1024² (development sources)
worst-case relative error (%)
ROM, span (linear rung)
R′=32
R′=128
R′=512
ROM, head
Transolver
DeepONet
U-Net
FNO
CG tol 0.03 (paper's FOM)
CG tolerance ladder
DST direct solve (exact to round-off)
0.33 ms, error ≈ 10⁻¹⁵ ↓
Heat 2D, 1024² (held-out)
R′=16
ROM, span (R′=32)
R′=48
R′=128
DeepONet
Transolver
U-Net
ROM, head R′=128
FNO
CN–CG ladder
CN–CG, dt 0.025, rtol 10⁻⁶ (paper's FOM)
modal (DST) solve = error reference
1.70 ms ↓
Burgers 2D, 1024² (development cases)
DeepONet
ROM, span
R′=32
ROM, head (R′=32...512)
U-Net
Transolver
FNO
Newton–BiCGStab ladder
R′=512
R′=384
pseudo-spectral Picard ladder (separate job, other GPU model)
reference solve 439 ms ↓
online time per query, GPU (ms)
Navier–Stokes 3D, 96³ (32 evaluation cases)
CNAB2, 10–50 steps: unstable ↑
DeepONet
ROM, span (16 modes)
U-Net
60 steps
Transolver
FNO
ROM, head (k = 8)
CNAB2, 70 steps (matched FOM)
reference (200 steps) 354 ms ↓
online time per query, GPU (ms)